



Twitter Stock Market Analysis

CoachX
LIVE

Twitter is one of the popular social media applications where people share what they feel in a limited number of words. Twitter is popular but not in the stock market. As Twitter is being delisted from the New York stock exchange, let's analyze the complete timeline of Twitter in the stock market.

Dataset Description

Twitter Stock Market Analysis Dataset:

- **Date:** The date on which the stock data is recorded.
- **Open:** The stock price at the opening of the trading day.
- **High:** The highest stock price during the trading day.
- **Low:** The lowest stock price during the trading day.
- **Close:** The stock price at the close of the trading day.
- **Adj Close:** The adjusted closing price accounting for any dividends or stock splits.
- **Volume:** The total number of shares traded during the day.

Task 1: Data Analysis

1. Read the data and display the first 100 rows from the data:

- This involves loading the dataset and showing the first 100 rows. This step helps in getting a preliminary view of the data and understanding its structure, including checking for data types and values.

2. Give the column insights:

- This task involves analyzing each column in the dataset to understand what kind of data it holds. Insights might include the range of values, data types, and any apparent patterns or anomalies.

3. Check whether this dataset contains any null values or not; if there are any, then remove the null values:

- This involves checking the dataset for missing or null values in any columns. If null values are present, this step would require handling them, typically by removing the rows with missing values or imputing them.

4. Find the statistical description of the data:

- This includes computing statistical metrics such as mean, median, standard deviation, minimum, and maximum values for numerical columns. This step provides a summary of the distribution and dispersion of the data.

5. Find the missing values in the data:

- Similar to checking for null values, this step specifically focuses on identifying any missing data points in the dataset. This involves quantifying the missing values and understanding their distribution across columns.

Task 2: Statistical Tests and Analysis (Optional)

1. Give me the Z-test or T-test over High, Low, and Close columns and see if the null hypothesis gets rejected or accepted:

- A Z-test or T-test is used to determine if there are significant differences between means of different groups. This involves comparing the means of the High, Low, and Close columns to check for any significant differences. The null hypothesis in these tests typically states that there is no significant difference between the groups being compared.

2. Check if the data is dependent or independent by using the chi-square method:

- The chi-square test is used to determine if there is a significant association between categorical variables. In this context, it might be used to test whether the data (e.g., stock prices) are dependent on some categorical factors. It assesses whether the observed frequencies differ from the expected frequencies under the null hypothesis of independence.

3. Write the insights which you have got from the above tests:

- This involves interpreting the results from the statistical tests conducted. Insights could include understanding whether there are significant differences or associations in the stock data and how these might impact the analysis.

Task 3: Visualization and Analysis

1. Show the Twitter stock prices over the years and give a conclusion:

- This task involves creating a visualization of the stock prices over time. It includes plotting stock prices across the years to identify trends, patterns, and overall movements. A conclusion would summarize the key findings from the visualization.

2. Compare the Close vs Date column for Twitter prices over the years:

- This involves creating a plot to compare the closing prices of Twitter stock with dates over time. The comparison aims to show how the closing price changes over time and identify any trends or patterns.

3. Assign buttons to control time periods. Add the buttons to analyze the stock prices of Twitter in different time periods:

- This task requires creating interactive visualizations where users can select different time periods to view the stock prices. Buttons or filters would allow users to analyze specific periods within the dataset.

4. Give the complete timeline of Twitter in the stock market (Line Graph):

- This involves creating a line graph that represents the entire timeline of Twitter's stock prices. The graph should show how the stock price has evolved over the entire period from its start in 2018 to the present.

5. Give the insights for the above analysis and make a word cloud for that analysis:

- This involves summarizing the findings from the visualizations and analysis. Insights might include key trends, patterns, or notable events. A word cloud can be created to visually represent the most frequently occurring words or phrases from the analysis.

6. Create a Dashboard from the above statements by using Power BI, Tableau, or Python and present it to the Client:

- This task involves compiling all the visualizations and analyses into a comprehensive dashboard using tools like Power BI, Tableau, or Python libraries. The dashboard should be designed to present the analysis clearly and interactively to the client.

Final Task: Summary Conclusion

Create a Summary Conclusion for the Entire Analysis and show the insights to the client:

- This involves summarizing the entire analysis, including key findings from the data analysis, statistical tests, visualizations, and dashboard. The summary should provide a clear and concise overview of the insights gained from the analysis and any recommendations or conclusions drawn from the data.

Advanced Questions(Optional) This section is for those who have little understanding of ML

1. How do seasonal patterns affect Twitter's stock prices, and can you identify specific months or quarters with significant fluctuations?

- **Hint:** Perform seasonal decomposition and analyze patterns using time series analysis.

2. What is the correlation between Twitter's stock price volatility and trading volume?

- **Hint:** Compute and visualize the correlation between volatility (standard deviation of prices) and trading volume.

3. How do Twitter's stock prices compare to other social media companies over the same period?

- **Hint:** Collect and analyze stock price data from similar companies for comparative analysis.

4. Can you predict Twitter's stock price movement using machine learning models?

- **Hint:** Apply regression models or time series forecasting methods to predict future stock prices.

5. How do major news events correlate with significant changes in Twitter's stock price?

- **Hint:** Conduct event study analysis to examine stock price movements around major news events.

6. What is the impact of market sentiment (derived from Twitter posts) on Twitter's stock prices?

- **Hint:** Analyze sentiment data from Twitter posts and correlate it with stock price movements.

7. How does the performance of Twitter's stock compare to the broader market indices during economic downturns?

- **Hint:** Compare Twitter's stock performance to market indices like S&P 500 during economic recessions.
8. **What are the patterns in the high and low stock prices and their impact on closing prices?**
- **Hint:** Analyze the relationship between high, low, and closing prices using regression analysis.
9. **How do stock price trends align with Twitter's financial performance indicators (e.g., earnings reports)?**
- **Hint:** Correlate stock price trends with financial performance data from earnings reports.
10. **What predictive features (technical indicators) provide the best forecasts for Twitter's stock prices?**
- **Hint:** Evaluate technical indicators like moving averages, RSI, and MACD for forecasting accuracy.

DOCUMENTATION

After Completion of the projects you have to Create one Docx file in that you have to Make Report of above Projects and include some Key Factors like

1. Introduction
 2. Methodology
 3. Requirement Analysis
 4. Other Parameters depending upon the Projects
 5. All Visualization like All Charts which is there in the Dashboards
 6. Insights from the Charts as well as Dashboards
 7. Conclusion
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After creating the Reports for above project upload that docx file or pdf file in Assignment links

Tip :- For your Reference One Sample report is associated with this Project . You can refer that project for your use-case .